INCH-POUND
MIL-S-24187/1A(SH)
22 March 1991
SUPERSEDING
MIL-S-24187/1(SH)
15 June 1978

MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL AND REMOTE OPERATION, STYLE LS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.

INCHES	MM
.001	0.03
.005	0.13
.006	0.15
.02	0.51
.06	1.52
.12	3.05
.28	7.11
.312	7.98
.33	8.38
.45	11.43
3.060	77.72
3.968	100.58
8.28	210.31
9.40	238.76
9.96	252.98
10.18	258.57
12.78	324.62

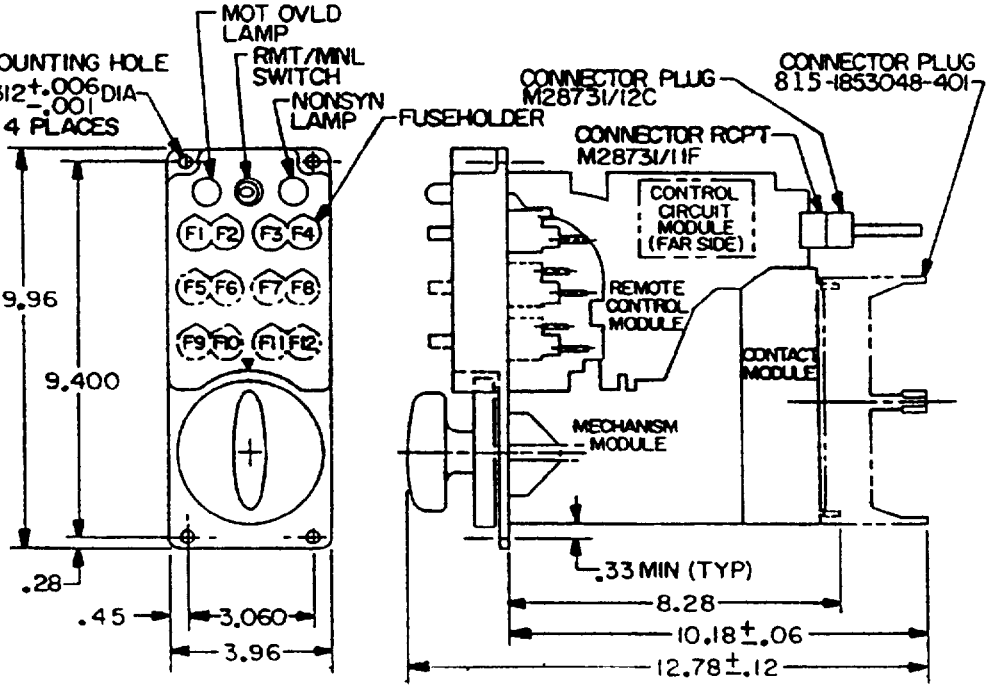


FIGURE 1. *Remote shown style LS switch assembly*

AMSC N/A FSC 5930

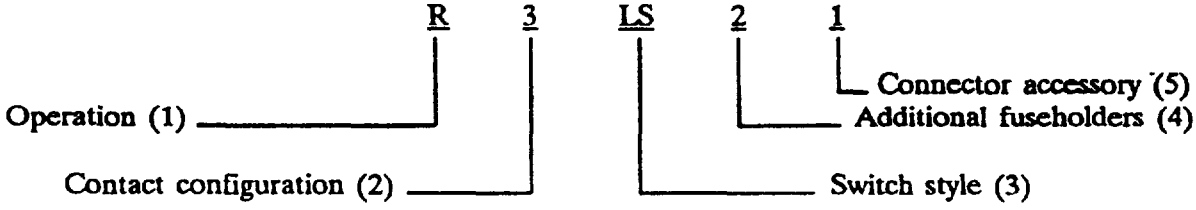
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NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are plus or minus 0.02 (0.51 mm) for two-place decimals and plus or minus 0.005 (0.13 mm) for three-place decimals.
3. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. *Remote shown style LS switch assembly – Continued.*

Type designation:



(1) Switch assemblies shall be remote operated with manual override, designated R, or manually operated, designated M.

(2) Contact configuration shall be type 2, 3, or 4 as specified (see figure 2).

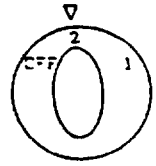
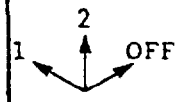
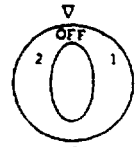
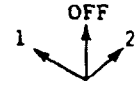
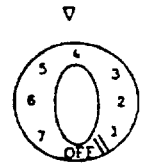
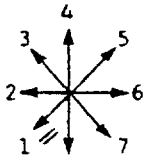
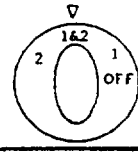
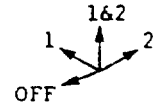
(3) Switch style shall be designated by the letters LS.

(4) Number of additional double circuit fuseholders installed shall be indicated by a single digit, 0 to 4 for remote operated switches and 0 to 6 for manually operated switches. Additional fuseholders are those provided for load circuit fuses. Two fuseholders, for fuses 1 through 4 (see figures 1 and 4) are required in remote operated switches for protection of control circuits.

(5) Connector accessory shall be provided when required (see figure 6).

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TABLE I. *Switch selection.*

Type designation				Application data					
Switch operation	Contact configuration	Switch style	Additional fuseholders	Switching action	Stop position[1]	Dial position[2]	Handle position[3]	Wired contacts[4]	Unwired contacts[4]
M R	2	LS	0-6 0-4	1-2-OFF	2&7			( )D LOAD 2 ( )C INPUT ( )B LOAD 1	( )E
M R	2	LS	0-6 0-4	1-OFF-2	2&7			( )E LOAD 2 ( )D-INPUT/OFF ( )C- ( )B LOAD 1	
M R	3	LS	0-6 0-4	1-4-7-OFF	4	 _5	 _5	HH,HE INPUT/OFF GE,GD LOAD 7 FD,FC LOAD 6 EC,EB LOAD 5 DB,DE LOAD 4 CE,CD LOAD 3 BD,BC LOAD 2 AC,AB LOAD 1	XB, XX
M R	4	LS	0-6 0-4	OFF-1-1&2-2	3&7			( )E LOAD 2 ( )D INPUT ( )C LOAD 1 ( )B OFF	

Example: Type designation for a manual switch assembly, with type 2 contact configuration, and six additional fuseholders is M2LS6.

[1]Stops shall be adjustable to limit switch handle rotation to the positions indicated or to any consecutive positions within the maximum rotation indicated.

[2]Reference position of an LS switch is considered to be that in which the flat on the handle shaft is up. The mark ∇ and the dial positions shown are used to designate the reference position

[3]Normal position of an LS switch is the position which coincides with normal circuit conditions in any application Handle shall be adjustable to point upward in the normal position or to any specified position, in 45 degree increments from upward. The handle and rotary description plate rotate

[4]See figures 2 and 3 for complete description of contact identification.

[5]Limit of rotation.

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REQUIREMENTS:

Construction: Style LS switch parts, modules, and assemblies shall be constructed in accordance with Drawing 815-1853048.

Interchangeability: Functional interchangeability of like modules shall be demonstrated by go-no-go tests in gauges fabricated in accordance with Drawing 803-1853220.

Characteristics: Characteristics shown in table II shall apply to the style LS switch assemblies.

TABLE II. *Characteristics*

Characteristics	Value		
	Contact configuration		
	2	3	4
Number of poles	56	24	56
Number of positions	2 or 3	2 to 8	4
Break-before-make	Yes	Yes	–[1]
Electrical rating and endurance loading			
10 Amperes, 115 V, 60 Hz, unity p.f.	20,000 Operations	15,000 Operations	20,000 Operations
7.5 Amperes, 115 V, 60 Hz, unity p.f	30,000 Operations	20,000 Operations	30,000 Operations
5 Amperes, 50 V, dc, resistive	30,000 Operations	20,000 Operations	30,000 Operations
2 Amperes, 115 V, dc, resistive	30,000 Operations	20,000 Operations	30,000 Operations
25 Microamperes, 1.5 V, dc, resistive	30,000 Operations	20,000 Operations	30,000 Operations
Remote operation.			
Transfer time			
Minimum	15 ms	10 ms	Not applicable[1]
Maximum	35 ms	35 ms	Not applicable[1]
Actuating time.			
1 Position change	0.25 sec	0.25 sec	0.25 sec
7 Position change (one direction)	–	1.85 sec	–

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TABLE II. *Characteristics – Continued.*

Characteristics	Value
Current drain: (At 115 V, 3Φ, 60 Hz)	1 ampere rms per phase
Contact resistance:	
At 10 amperes	16 milliohm
At 25 microamperes	50 milliohm
Torque:	
Shaft rotation	
Type RLS	50 pound-inches
Type MLS	40 pound-inches
Stop strength	100 pound-inches
Weight:	
Type RLS	14 pounds
Type MLS	8 pounds
Temperature and humidity (for rotational torque and endurance tests):	
Minimum	Minus 20 °F (minus 29 °C)
Maximum	Plus 149 °F (plus 65 °C) and 40 to 50 percent rh
Shock	MIL-S-901, grade A, class 1, type C
Vibration	MIL-STD-167-1, type I, except variable frequency
Temperature rise	30 °C

[1]Switch is a combination of make-before-break (MBB) and break-before-make (BBM) MBB is provided between 1 or 2 and 1 and 2 positions. All other functions are BBM (see figure 2).

Switch assembly modules: Manual operated LS switch assembly shall consist of the following modules as detailed on Drawing 815-1853048:

Front module
Mechanism module
Switch contact module

Remote operated LS switch assembly shall have, in addition to the above, the following modules:

Remote control module
Control circuit module.

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Switch function: Style LS switch assemblies shall provide the switching functions as shown on figure 2. Contact configuration 2 shall provide 56 two- or three-position poles, each consisting of 4 stationary contacts B, C, D, and E and a movable contact. Contact configuration 3 shall provide 24 two- to eight-position poles, each consisting of 9 stationary contacts X, A, B, C, D, E, F, G, and H and a movable contact. Contact configuration 4 shall provide 56 four-position poles, each consisting of 3 stationary contacts C, D, and E and a movable contact. Stops shall prevent transfer of the movable contacts into unused positions. Stops (removable) shall be provided which, when inserted as specified in table I, will limit movement to consecutive used positions. Stops shall be as detailed on Drawing 815-1853048 and shall be accessible from the front of the switch assembly, after removal of the handle and rotary description plate.

Contact designation: Each stationary switching contact shall be identified by two letters and a number. The first letter shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figure 2. The number shall be that of the column in which the contact is located (see figure 3). Each contact in the switch contact module connector plug shall be assigned the same identification as the contact in the switch contact module to which it connects.

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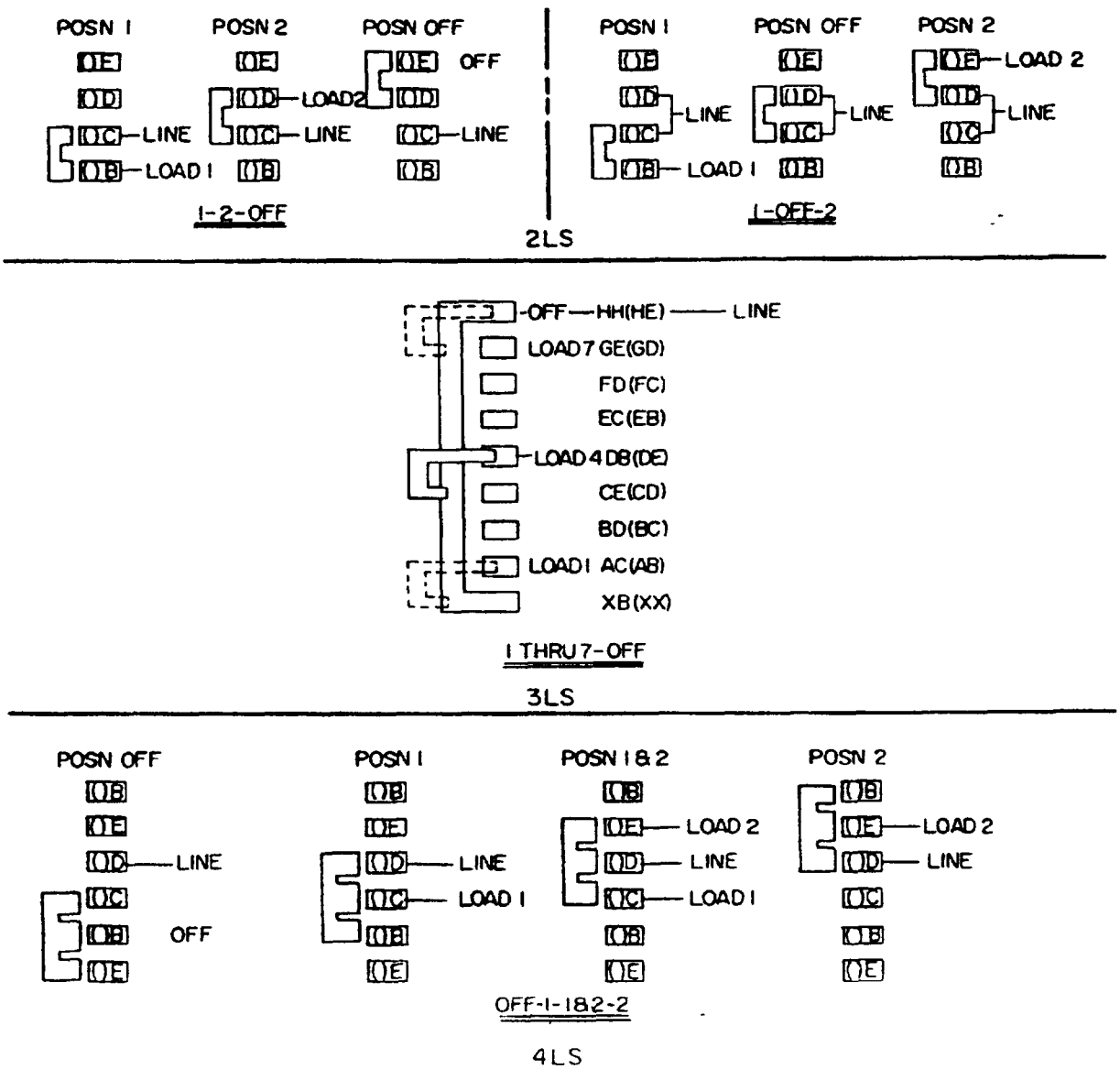
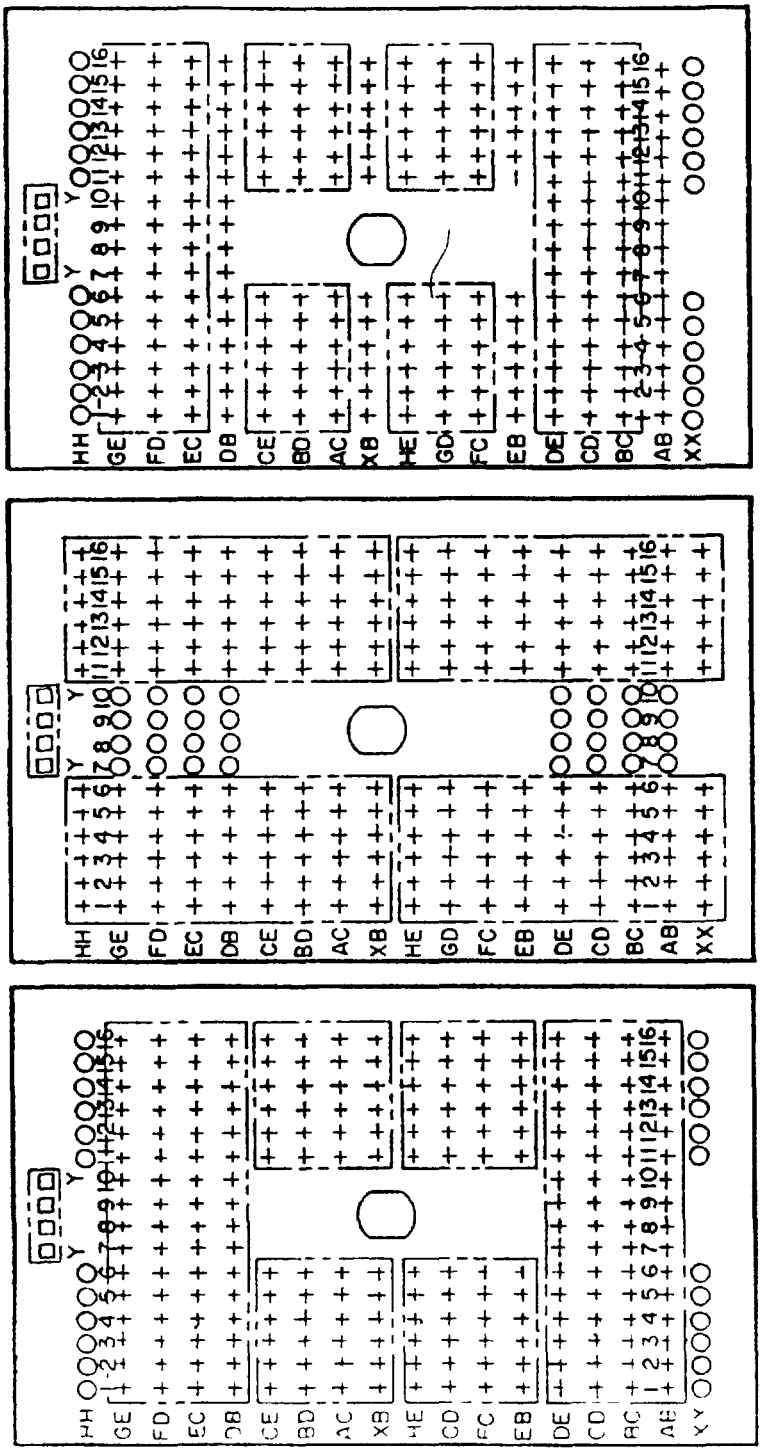


FIGURE 2 *Contact configuration*

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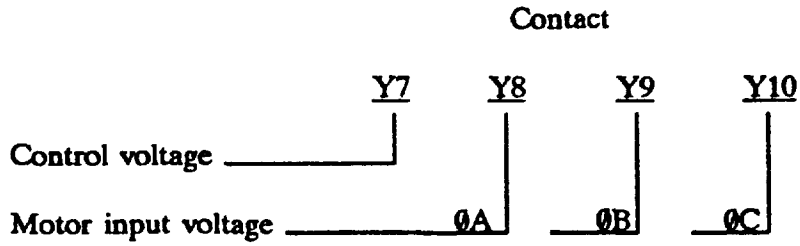


NOTE Bushings and contacts shall be installed in the stationary plates (815-1853048-301) as indicated. Crosses signify bushings and contacts which are installed. Circles signify omitted bushings and contacts. Squares show locations of bushings and contacts required for remote operation. Contact location markings and dashed rectangles, enclosing groups of poles, are shown for information only. Contact location markings shall be molded onto the front and rear surfaces of connector plugs (815 1853048-401).

FIGURE 3. *Stationary plate – rear view.*

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Remote operation: Connections to the switch contact module for remote operation shall be as follows:



Remote operated switch assemblies shall move to the positions shown in table III when the specified control voltage is applied.

TABLE III. *Control voltage for remote operation.*

Control voltage	Closed switch contacts				XFMR TAP [1]
	2LS		3LS	4LS	
	1-OFF-2	1-2-OFF			
0	–	–	X-A	–	A
16 4	–	–	X-B	B-C	B
32.8	B-C	B-C	X-C	B-C-D	C
49.2	C-D	C-D	X-D	C-D-E	D
65.6	D-E	D-E	X-E	D-E	E
82.0	–	–	X-F	–	F
98 4	–	–	X-G	–	G
114 8	–	–	X-H	–	H

[1]See figure 3 of MIL-S-24187 for typical test circuit for remote operated switch assemblies.

Fuseholder connector wiring, 38-pin receptacle M28731/11F (815-1853048-58): Wiring shall be as shown in tables IV and V.

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TABLE IV. *Fuse connections*

Fuse number	Contact of 38 pin connector	
	Line	Load
1[1]	D	K
2[1]	C	J
3[1]	B	F
4[1]	A	E
5	P	U
6	N	T
7	M	S
8	L	R
9	W	Y
10	BB	HH
11	AA	DD
12	V	X

[1]Fuseholders for fuses 1 through 4 (for remote operated switch assemblies) shall be wired through the 20 contact connector (815-1853048-65 and 815-1853048-136) Remote control circuit shall be as shown on figure 4

TABLE V. *Other connections.*

Contact of 38 pin connector	Circuit
TT	Supply, blown fuse indicating[1]
PP	Return, blown fuse indicating[1]
NN	Switch in manual[2]
JJ	Designation switch active[2]
KK	Designation switch active[2]

[1]Fuseholders shall be wired in parallel
[2]Remote operated switch assemblies only (see figure 4)

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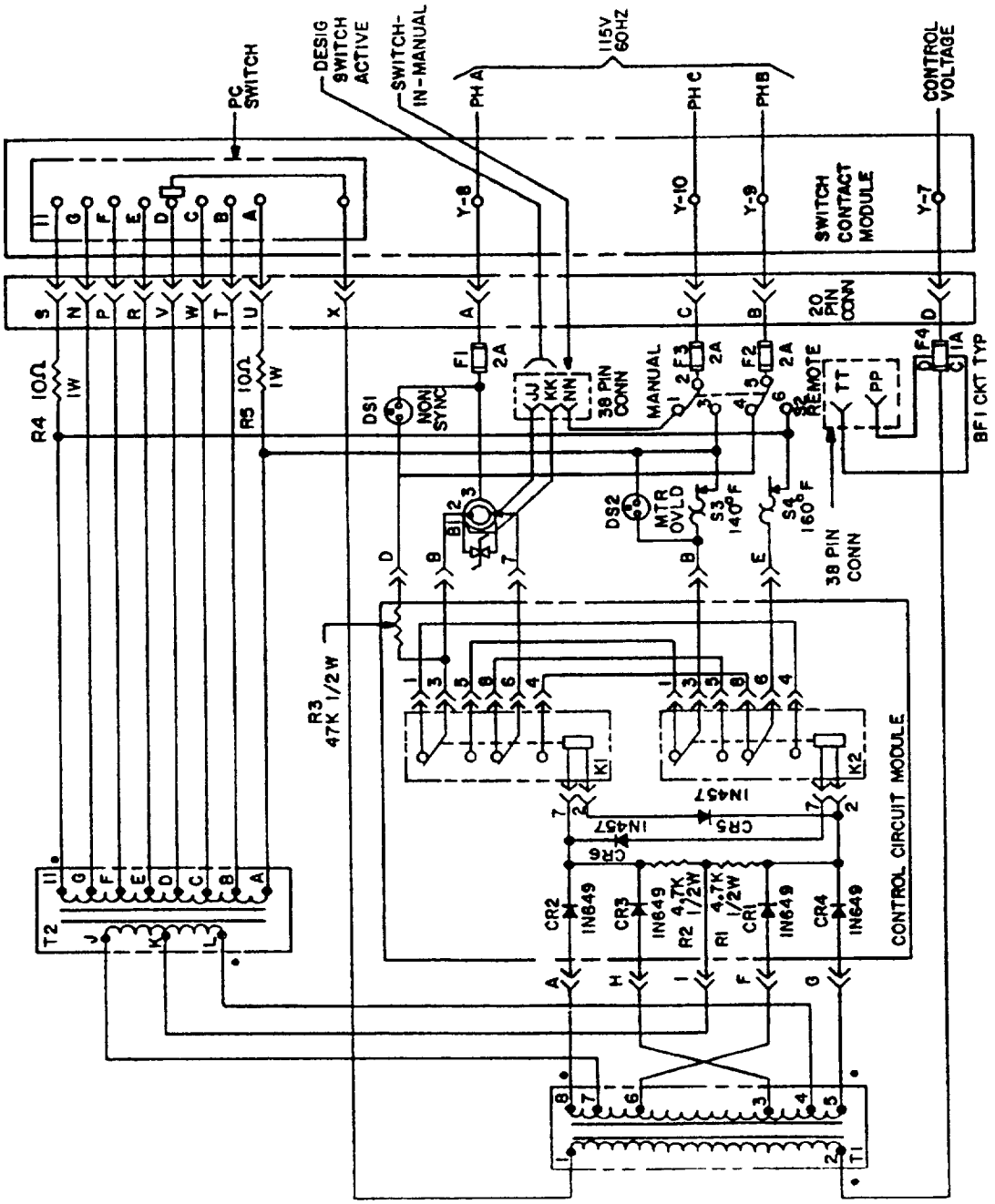


FIGURE 4. *Remote control circuit.*

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Repair parts: Repair parts shall be as specified in table VII.

TABLE VII. *Repair parts*

Drawing 815-1853048 piece No.		Description	Quantities required for quantity of switches installed					
			1–3	4–12	13–30	31–60	61–120	120 or more
Manual	Remote							
	1	Switch, type R2LS4	1	1	1	1	2	3
	2	Switch, type R3LS4	1	1	1	1	2	3
	3	Switch, type R4LS4	1	1	1	1	2	3
4		Switch, type M2LS6	1	1	1	1	2	3
5		Switch, type M3LS6	1	1	1	1	2	3
6		Switch, type M4LS6	1	1	1	1	2	3
35	35	Stop screw	1	1	1	2	2	3
41	41	Detent spring	1	1	1	2	3	6
	62	Toggle switch	1	1	1	2	3	4
	59	Indicator light	1	1	1	2	3	5
	60	Indicator light	1	1	1	2	3	5
		Lamp (M15098/11-001)	Note 2					
	151	Control circuit module	1	1	1	2	2	3
	154	Relay	1	2	3	Note 3		
	163	Washer, insul	2	2	2	4	6	8
	164	Spacer, insul.	2	2	2	4	6	8
	56	Fuseholder	1	1	1	2	3	4
		Fuse carrier (1A0981)	1	1	2	3	4	5
	57	Fuse, 2.0 ampere	Note 4					
	166	Fuse, 1.0 ampere	Note 4					
A/R	A/R	Fuse, load circuit	Note 4					

NOTES

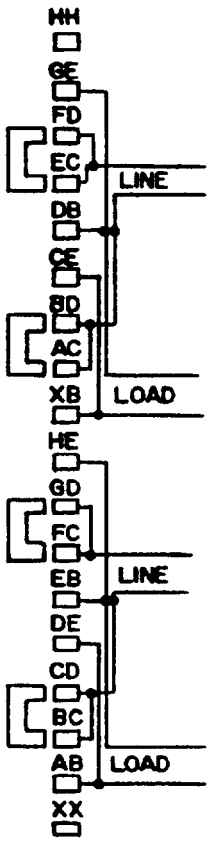
1. Refer to Drawing 803-4680149 for piece numbers.
2. One hundred percent repair lamps shall be provided
3. Supply one for each 10 or fraction of 10 installed.
4. One hundred fifty percent repair fuses shall be supplied Fuses shall be supplied in multiples of five, in standard commercial packages, with a minimum quantity of 15

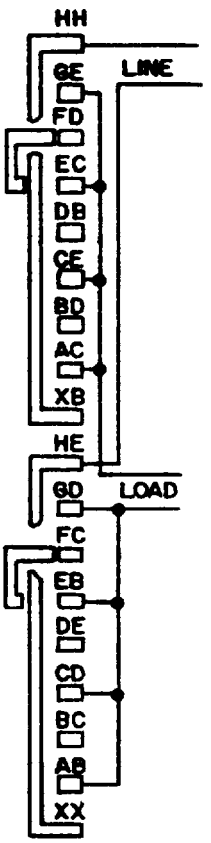
Qualification samples — Two remote operated switch assemblies of each contact configuration shall be submitted for qualification inspection

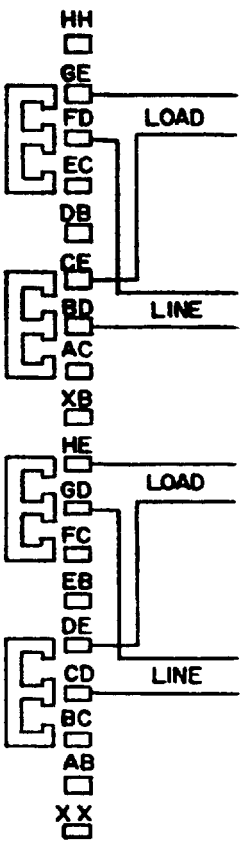
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TESTS:

Loading of switches: Loading of switches for endurance and temperature rise tests for qualification and quality conformance, group C testing, shall be in accordance with figure 5 and table VII.







TYPE 2LS

COLUMNS	VOLTS	AMPS
1	115	10AC
5	50	5DC
7 & 9	115	7.5AC
11	115	2DC
16	1.5	25μDC

TYPE 3LS

COLUMNS	VOLTS	AMPS
1 & 2	115	10AC
5 & 6	50	5DC
11 & 12	115	2DC
16	1.5	25μDC
13 & 14	115	7.5AC

TYPE 4LS

COLUMNS	VOLTS	AMPS
1	115	10AC
5	50	5DC
11	115	2DC
16	1.5	25μDC
7 & 9	115	7.5AC

FIGURE 5 *Test loading*

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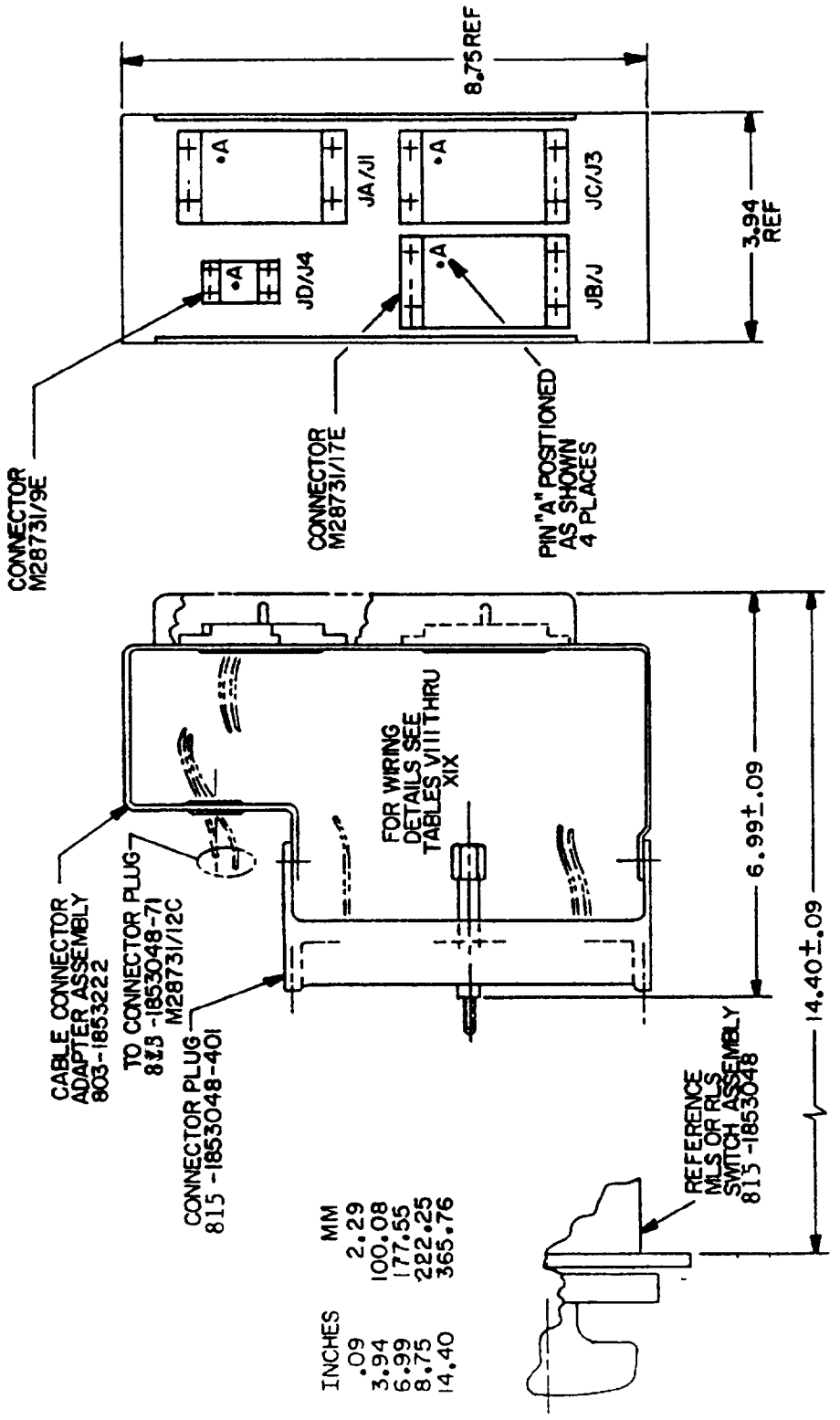
Temperature rise: Contacts to be monitored during temperature rise test are listed in table VII.

TABLE VII. *Temperature measurements.*

Column	Contacts		
	Type 2LS	Type 3LS	Type 4LS
1 5	EC, DB EC, DB	HH, GE HH, GE	GE, FD GE, FD

Connector accessories: When specified in the contract or order, switch assemblies shall be furnished with connector accessories in accordance with figure 6. Connector accessories shall be wired in accordance with tables VIII through XIX, as applicable.

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Note
[1]For applications subject to mechanical shock and vibration, rear support shall be provided by the installing activity.

FIGURE 6. *Connector accessory for connection of LS switch to ship's cables.*

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Wiring connections for connector accessory:

Each connector (JA/J1 through JP/J4) shall be wired to a specific group of connector plugs (pcs. 815-1853048-71 or 815-1853048-401) using the data shown in tables VIII through XIX. Wire color coding shall be as follows:

JA/J1 – Black
JB/J2 – White
JC/J3 – Red
JD/J4 – White

TABLE VIII. *Wiring connections, for connector JA/J1 for 2LS contact configuration – switching actions 1, 2, off.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
FC LINE INPUT	CC	CM	CB	CL	CA	CK					CH	BZ	CF	BY	CE	BX
EC LINE INPUT	AN	AX	AM	AW	AL	AV	AK	AU	AJ	AT	AH	AS	AF	AR	AE	AP
BC LINE INPUT	DH	CX	DF	CW	DE	CV	DD	CU	DC	CT	DB	CS	DA	CR	CZ	CP
AC LINE INPUT	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
	Contacts in 120 pin connector															

TABLE IX. *Wiring connections, for connector JA/J1 for 2LS contact configuration – switching action1, off, 2.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
GD[1]	CC	CM	CB	CL	CA	CK					CH	BZ	CF	BY	CE	BX
FC																
FD	AN	AX	AM	AW	AL	AV	AK	AU	AJ	AT	AH	AS	AF	AR	AE	AP
EC																
CD	DH	CX	DF	CW	DE	CV	DD	CU	DC	CT	DB	CS	DA	CR	CZ	CP
BC																
BD	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
AC																
	Contacts in 120 pin connector															

[1]Install jumpers from contact rows GD to FC, FD to EC, CD to BC, and BD to AC

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TABLE X. *Wiring connections, for connector JB/J2 for 2LS contact configuration – switching actions1, 2, off and 1, off, 2.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
XB-LOAD 1	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
EB-LOAD 1	CC	CM	CB	CL	CA	CK					CH	BZ	CF	BY	CE	BX
DB-LOAD 1	AN	AX	AM	AW	AL	AV	AK	AU	AJ	AT	AH	AS	AF	AR	AE	AP
AB-LOAD 1	DH	CX	DF	CW	DE	CV	DD	CU	DC	CT	DB	CS	DA	CR	CZ	CP
	Contacts in 120 pin connector															

TABLE XI. *Wiring connections, for connector JC/J3 for 2LS contact configuration – switching actions 1, 2, off.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
GD-LOAD 2	CC	CM	CB	CL	CA	CK					CH	BZ	CF	BY	CE	BX
FD-LOAD 2	AN	AX	AM	AW	AL	AV	AK	AU	AJ	AT	AH	AS	AF	AR	AE	AP
CD-LOAD 2	DH	CX	DF	CW	DE	CV	DD	CU	DC	CT	DB	CS	DA	CR	CZ	CP
BD-LOAD 2	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
	Contacts in 120 pin connector															

TABLE XII. *Wiring connections, for connector JC/J3 for 2LS contact configuration – switching actions 1, off, 2*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
HE-LOAD 2	CC	CM	CB	CL	CA	CK					CH	BZ	CF	BY	CE	BX
GE-LOAD 2	AN	AX	AM	AW	AL	AV	AK	AU	AJ	AT	AH	AS	AF	AR	AE	AP
DE-LOAD 2	DH	CX	DF	CW	DE	CV	DD	CU	DC	CT	DB	CS	DA	CR	CZ	CP
CE-LOAD 2	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
	Contacts in 120 pin connector															

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TABLE XIII. *Wiring connections, for connector JA/J1 for 3LS contact configuration.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
_[1]																
XX LINE INPUT	CC	BV	CB	BU	CA	CK					CH	BZ	BT	BY	BS	BX
XB LINE INPUT	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
Contacts in 120 pin connector																

[1]In this application the XX and XB contacts are input (line) vice HE and HH

TABLE XIV. *Wiring connections, for connector JB/J2 for 3LS contact configuration.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
EC-LOAD 5	AC	AB	AL	AA	AK	Z					AJ	Y	AH	X	W	V
EB-LOAD 5	CX	CM	CW	CL	CV	CU					CT	CS	CF	CR	CE	CP
CE-LOAD 3	AN	AX	AM	AW	AV	AU					AT	AS	AF	AR	AE	AP
CD-LOAD 3	DH	DS	DF	DR	DE	DD					DC	DB	DA	DK	CZ	DJ
AC-LOAD 1	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
AB-LOAD 1	EB	EA	DZ	DP	DY	DN					DX	DM	DW	DL	DV	DU
GD-LOAD 7	CC	BV	CB	BU	CA	CK					CH	BZ	BT	BY	BS	BX
GE-LOAD 7	U	J	T	H	S	F					R	P	N	M	B	L
Contacts in 120 pin connector																

TABLE XV. *Wiring connections, for connector JC/J3 for 3LS contact configuration.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
DE-LOAD 4	CX	CM	CW	CL	CV	CU					CT	CS	CF	CR	CE	CP
DB-LOAD 4	AN	AX	AM	AW	AV	AU					AT	AS	AF	AR	AE	AP
BD-LOAD 2	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
BC-LOAD 2	DH	DS	DF	DR	DE	DD					DC	DB	DA	DK	CZ	DJ
FC-LOAD 6	CC	BV	CB	BU	CA	CK					CH	BZ	BT	BY	BS	BX
FD-LOAD 6	AC	AB	AL	AA	AK	Z					AJ	Y	AH	X	W	V
Contacts in 120 pin connector																

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TABLE XVI *Wiring connections, for connector JA/J1 for 4LS contact configuration.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
GD LINE INPUT	CC	CM	CB	CL	CA	CK					CH	BZ	CF	BY	CE	BX
FD LINE INPUT	AN	AX	AM	AW	AL	AV	AK	AU	AJ	AT	AH	AS	AF	AR	AE	AP
CD LINE INPUT	DH	CX	DF	CW	DE	CV	DD	CU	DC	CT	DB	CS	DA	CR	CZ	CP
BD LINE INPUT	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
	Contacts in 120 pin connector															

TABLE XVII *Wiring connections, for connector JB/J2 for 4LS contact configuration.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
FC-LOAD 1	CC	CM	CB	CL	CA	CK					CH	BZ	CF	BY	CE	BX
EC-LOAD 1	AN	AX	AM	AW	AL	AV	AK	AU	AJ	AT	AH	AS	AF	AR	AE	AP
BC-LOAD 1	DH	CX	DF	CW	DE	CV	DD	CU	DC	CT	DB	CS	DA	CR	CZ	CP
AC-LOAD 1	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
	Contacts in 120 pin connector															

TABLE XVIII. *Wiring connections, for connector JC/J3 for 4LS contact configuration.*

Contacts in connector plug 815-1853048-401																
Column / Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
HE-LOAD 2	CC	CM	CB	CL	CA	CK					CH	BZ	CF	BY	CE	BX
GE-LOAD 2	AN	AX	AM	AW	AL	AV	AK	AU	AJ	AT	AH	AS	AF	AR	AE	AP
DE-LOAD 2	DH	CX	DF	CW	DE	CV	DD	CU	DC	CT	DB	CS	DA	CR	CZ	CP
CE-LOAD 2	BP	BF	BN	BE	BM	BD					BB	BL	BA	BK	AZ	BJ
	Contacts in 120 pin connector															

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TABLE XIX. *Wiring connections for connector JD/J4.*

20 Pin connector JD/J4 contacts	Connector plug contacts 815-1853048		Service
	401	71	
A	Y8		Phase A
B	Y9		Phase B
C	Y10		Phase C
D	Y7		Control
T		NN	Switch-in-manual
U		JJ	Designation switch active
V		KK	Designation switch active
W		PP	BFI return
X		TT	BFI supply

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 5930-N681-02)

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